2 An executive toy consists of five identical steel spheres of mass m suspended so that they are free to move in a vertical plane as shown in Fig. 2.1. Each sphere is suspended using thin inextensible strings of negligible mass.

You may assume that collisions between the spheres are perfectly elastic and air resistance is negligible

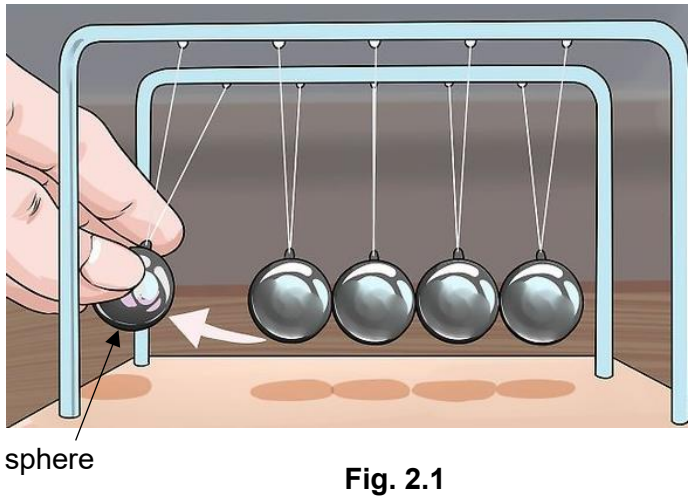

Fig. 2.1

(a) The first sphere is displaced to the left and then released. Describe and explain the resulting motion of the five spheres.

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……[3]

(b) **(i)** Object **A** has a velocity of u and moves horizontally towards Object **B** which is at rest. The collision between both objects is head-on and perfectly elastic.
The ratio of the mass of Object **A** to the mass of Object **B** is **1:2**.
Determine the ratio of the **final** velocity of Object **B** to the initial velocity of Object **A**.

ratio = ……..[2]

(b) **(ii)** Object **C** has a velocity of u and moves horizontally towards Object **D** which is at rest. The collision between both objects is head-on and perfectly elastic.
The ratio of the mass of Object **C** to the mass of Object **D** is **2:1**.
Determine the ratio of the final velocity of Object **D** to the initial velocity of Object **C**.

ratio = ……..[1]

(iii) Hence, if the **<u>second and fourth spheres are replaced with steel spheres of mass 2*m*</u>** of the executive toy in Fig. 2.1, determine the ratio of

$$\frac{\text{Maximum vertical displacement of the first sphere}}{\text{Maximum vertical displacement of the last sphere}}$$

Assume that the collision between both objects is head-on and perfectly elastic.

ratio = ……..[2]

(c) The five identical steel spheres in Fig. 2.1 are of mass m.

State and explain one significant difference to the motion observed if all the thin inextensible strings are replaced by springs of negligible mass.

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……[2]